

Routing games

1. A Nash flow is a flow in which ...
 - (a) the total cost is minimised
 - (b) every path carries equal flow
 - (c) all traffic uses a single path
 - (d) no individual can reduce their own cost by switching to another path
2. An optimal flow is the flow that ...
 - (a) equals the Nash flow in every game
 - (b) uses the most paths
 - (c) no driver can improve on individually
 - (d) minimises the total cost across all traffic
3. The price of anarchy measures ...
 - (a) the number of paths in the network
 - (b) the cost of the cheapest path
 - (c) the ratio of the Nash flow cost to the optimal flow cost
 - (d) how many drivers there are
4. Braess's paradox is the observation that ...
 - (a) the Nash flow always equals the optimal flow
 - (b) removing a road always helps
 - (c) adding a new road can make every driver's commute worse
 - (d) drivers never reach an equilibrium
5. A feasible flow that minimises the potential function Φ is ...
 - (a) the optimal flow
 - (b) a flow that uses no congested edges
 - (c) never an equilibrium
 - (d) a Nash flow
6. The optimal flow of a routing game equals the Nash flow of the game with ...
 - (a) all costs set to zero
 - (b) the costs doubled
 - (c) each cost function replaced by its marginal cost
 - (d) a single combined path

7. A feasible flow is one in which ...
 - (a) the total cost is minimised
 - (b) the flow on the paths of each commodity sums to its demand
 - (c) no path is used
 - (d) all flow uses a single path
8. Edge cost (latency) functions in a routing game are assumed to be ...
 - (a) negative
 - (b) constant
 - (c) non-negative, continuous and non-decreasing in the flow
 - (d) decreasing in the flow
9. The price of anarchy is always ...
 - (a) at least 1
 - (b) negative
 - (c) exactly 1
 - (d) at most 1
10. In Pigou's example (one fixed-cost route and one congestible route) the price of anarchy is ...
 - (a) 1
 - (b) $4/3$
 - (c) $8/7$
 - (d) 2
11. Braess's paradox arises because each driver ...
 - (a) cannot find the shortest path
 - (b) is forced onto a single route
 - (c) pays a toll
 - (d) ignores the extra congestion they impose on others (an externality)
12. At a Nash flow, all paths a commodity actually uses ...
 - (a) are the cheapest in the whole network
 - (b) carry equal flow
 - (c) have different costs
 - (d) have equal (and minimal) cost
13. The total cost of a flow is ...

- (a) $\sum_e c_e(f_e)f_e$, summed over the edges
 - (b) the cost of the cheapest edge
 - (c) the price of anarchy
 - (d) the number of paths used
14. The optimal flow is found by ...
- (a) maximising the total cost
 - (b) letting each driver choose selfishly
 - (c) using only one path
 - (d) minimising the total cost over all feasible flows
15. Marginal-cost pricing aligns the selfish and social outcomes because ...
- (a) it sets all costs to zero
 - (b) it forbids the use of congested edges
 - (c) it removes all congestion
 - (d) it makes each driver internalise the congestion they cause
16. A routing game is written (G, r, c) , where r denotes the ...
- (a) number of edges in the network
 - (b) price of anarchy
 - (c) reciprocal of the cost
 - (d) amount of traffic (demand) each source-sink commodity must route
17. The potential function of a routing game is ...
- (a) $\Phi(f) = \sum_e \int_0^{f_e} c_e(x) dx$
 - (b) $\Phi(f) = \sum_e f_e$
 - (c) $\Phi(f) = \max_e c_e(f_e)$
 - (d) $\Phi(f) = \sum_e c_e(f_e)f_e$
18. The marginal cost of an edge with cost function c is ...
- (a) $c^*(x) = \int_0^x c(t) dt$
 - (b) $c^*(x) = c(x)/x$
 - (c) $c^*(x) = \frac{d}{dx}(x c(x))$
 - (d) $c^*(x) = c'(x)$
19. The cost of a path P at flow x is ...
- (a) the number of edges it contains

- (b) the sum of the edge costs along it, $c_P(x) = \sum_{e \in P} c_e(x)$
 - (c) the cost of its cheapest edge
 - (d) the cost of its most expensive edge
20. In Braess's paradox, adding a zero-cost road changes the Nash flow so that its cost ...
- (a) rises, even though a new option was added
 - (b) becomes zero
 - (c) always falls
 - (d) stays exactly the same